EE-312L Signals and Systems Lab
Laboratory Session 09

To Understand and Demonstrate the Continuous-Time Fourier Transform Using MATLAB

December 4, 2024

1 Objective

The objective of this lab session is to understand and demonstrate the continuous-Time Fourier Transform (CTFT) using MATLAB.

2 Continuous-Time Fourier Transform (CTFT)

A CT signal $x(t)$ and its frequency domain, Fourier transform signal, $X(j\omega)$, are related by

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \tag{1}$$

where $\omega = 2\pi f$, ω is measured in rad/s and f is measured in Hz. The $X(j\omega)$ in (1), is referred to as **Analysis Equation**.

To find the inverse FT, we use the **Synthesis Equation**, given in (1)

$$x(t) = \int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega. \tag{2}$$

3 Continuous-Time Fourier Transform Using MATLAB

The following command can be used to compute Fourier Transform of a signal

$fourier(f)$: It returns the Fourier Transform of f. By default, the function *symvar* determines the independent variable, and ω is the transformation variable.

$fourier(f, transVar)$: It uses the transformation variable $transVar$ instead of ω.

$fourier(f, var, transVar)$: It uses the independent variable var and the transformation variable $transVar$ instead of $symvar$ and ω, respectively.

Example 01:

Find the Fourier Transform of $x(t) = e^{-2t^2}$

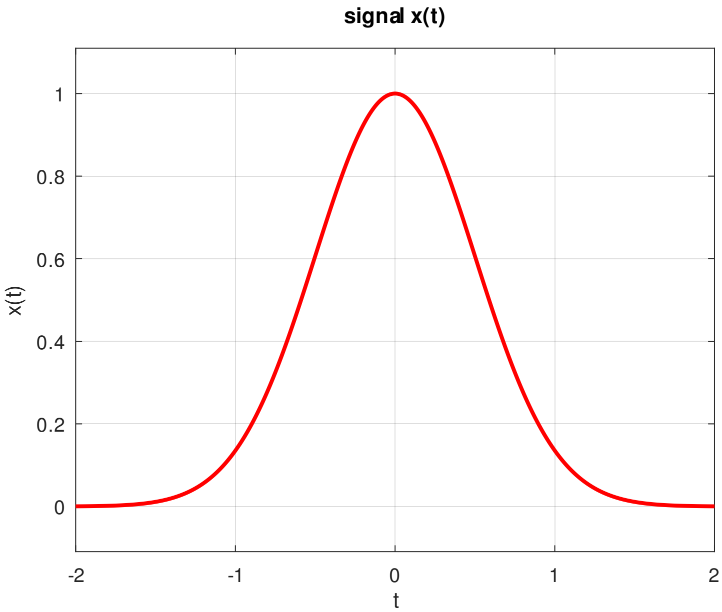

Figure 1: Continuous-time signal $x(t)$

```
clc;
clear all;
close all;
syms t;
figure(1)
x = exp(-2*t^2)
h1 = ezplot(x,[-2 2])
title('signal x(t)')
xlabel('t')
ylabel('x(t)')
set(h1, 'Color', 'r','LineWidth',2);
grid on;

X = fourier(x)
figure(2)
h2=ezplot(X)
title('Fourier Transform of x(t)')
```

```
xlabel('\omega')
ylabel('x(j\omega)')
set(h2, 'Color', 'r','LineWidth',2);
grid on;
```

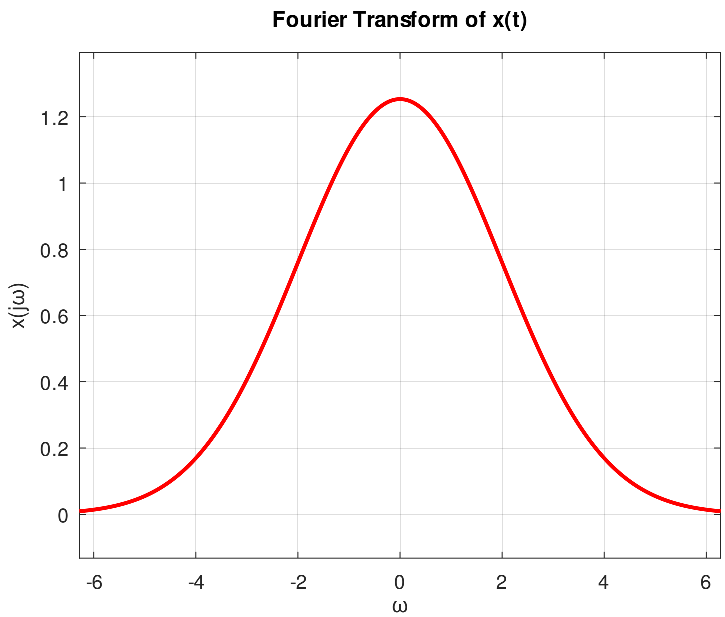

Figure 2: Fourier Transform $X(j\omega)$

Example 2:

Find the Fourier Transform of $x(t) = e^{-t^2}$

```
clc;
clear all;
close all;
syms t w
x = exp(-t^2);
X = fourier(x)
```

Answer : $\sqrt{\pi}e^{\frac{\omega^2}{4}}$

Example 3:

Find the Fourier Transform of $x(t) = e^{-t}u(t)$

```
clc;
clear all;
close all;
```

```
syms t w
x = exp(-t)*heaviside(t);
X = fourier(x)
```

Answer : $\frac{1}{1+j\omega}$

Example 4:

Find the Fourier Transform of $x(t) = e^{-2t}u(t)$

```
clc;
clear all;
close all;
syms t w
x = exp(-2*t)*heaviside(t);
X = fourier(x)
```

Answer : $\frac{1}{2+j\omega}$

4 Lab 09: Lab Tasks

Compute the Fourier Transform of the following signals using MATLAB/OCTAVE

1. $x_1(t) = e^{-5t}u(t)$
2. $x_2(t) = te^{-5t}u(t)$